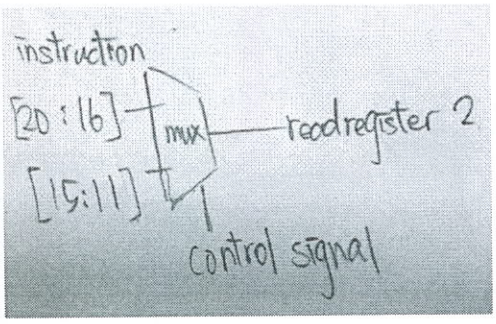
1. max：sign=0, exponent=11110, fraction=1111111111 => max=$1.1111111111 \times 2^{\cdots}$ $=2^{\cdots} \times (2 - 2^{-10})$=65504(D)
   min：sign=0, exponent=00001, fraction=0000000000 => min=$1 \times 2^{1-15}=2^{-14}$=0.00006103515625(D)
2. $4.12109375*10^{-1}$=0.0110100110 => exponent=13, fraction=1010011000
   $3.45703125*10^{-1}$=0.0101100010 => exponent=13, fraction=0110001000
   => $4.12109375*10^{-1}+3.45703125*10^{-1}$=0.1100001000 => exponent=14, fraction=1000010000
   $2.143*10^{3}$=100001011111 => exponent=26, fraction=0000101111
   => $1.1000010000*2^{-1}+1.000\ 0101\ 1111*2^{11}$
   => $0.000\ 000\ 000\ 0011000010000*2^{11}+1.000\ 0101\ 1111*2^{11}$
   (grs)
   =$1.000\ 0101\ 1111*2^{11}$(B) round to nearest even =>$1.000\ 0101\ 1111*2^{11}$(B) =2143(D)
   (grs)
3. $3.45703125*10^{-1}$=0.0101100010 => exponent=13, fraction=0110001000
   $2.143*10^{3}$=100001011111 => exponent=26, fraction=00001|1111
   => $1.01100010*2^{-2}+1.00001\ 1111*2^{11}$
   => $0.000\ 000\ 000\ 000\ 101100010*2^{11}+1.00001\ 1111*2^{11}$
   gr s
   =$1.000\ 0101\ 1111*2^{11}$ round to nearest even =>$1.000\ 0101\ 1111*2^{11}$
   grs
   $4.12109375*10^{-1}$=0.0110100110 => exponent=13, fraction=1010011000
   $1.000\ 0101\ 1111*2^{11}+1.101\ 001\ 1000*2^{-2}$
   =>$1.000\ 0101\ 1111*2^{11}+0.000\ 000\ 000\ 000\ 110100110*2^{11}$
   gr s
   =$1.000\ 0101\ 1111*2^{11}$ round to nearest even =>$1.000\ 0101\ 1111*2^{11}$(B)=2143(D)
   grs
4. LWI Rt, Rd(Rs) => Reg[Rt]= Mem[Reg[Rs]+Reg[Rd]]
   (a) PC, instruction memory, registers, ALU, data memory, Add, Mux
   since {RegWrite, MemRead, ALUSrc, MemWrite, ALUop, MemtoReg, Branch}={1, 1, 0, 0, 00, 1, 0}
   (b) Deploy a MUX ahead of register with output terminal to Read_Register_2, and input terminal to instruction[20:16] and instruction[15:11], so as to read Rs and Rd simultaneously and write data to Rt.
   (c) A 1-bit signal to the additional MUX is required to decide whether to read Rt or Rd



5. addi Rt, Rs, 40

(a) {RegWrite, MemRead, ALUSrc, MemWrite, ALUop, MemtoReg, Branch}={1, 0, 1, 0, 00, 0, 0} + {RegDst} + {0}

(b) ALU

(c) Add, Data memory

6.

I-Mem Read/ Write	Add	Mux	ALU	Regs Read/ Write	D-Mem Read/ Write	Sign-Extend	Sift-left-2
240ps	50ps	40ps	120ps	120ps	280ps	30ps	20ps

(a) I-Mem => 240ps

(b) I-Mem+Regs+Mux+ALU+Mux+Regs=240+120+40+120+40+120=680ps

(c) branch, Jump instruction;

branch：I-Mem+Regs+Mux+ALU+Mux=240+120+40+120+40=560ps

Jump：I-Mem+Shift-left-2+Mux=240+20+40=300ps

(d) I-Mem+Regs +ALU+D-Mem+Mux+Regs=240+120 +120+280+40+120=920ps

(e) load